1.1 First steps

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This unit outlines why R is worth learning for corpus-linguistic research and guides you through installing R and RStudio.

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Why learn R to begin with?

When it comes to data analysis, learning R offers an overwhelming number of short- and long-term advantages over conventional spreadsheet software such as Microsoft Excel or LibreOffice Calc:

- First of all, it's completely **free**. There's no need to obtain any expensive licences, as is the case for commercial software such as SPSS or MS Excel.
- R makes it very easy to document and share every step of the analysis, thereby facilitating **reproducible workflows**.
- Large (and by that I mean extremely large!) datasets pose no problems whatsoever. Loading tabular data with hundreds of thousands (or even millions) of rows takes only a few seconds, whereas most other software would crash.
- There are numerous extensions that provide tailored **functions for corpus linguistics**, which aren't available in general-purpose spreadsheet software. These allow us to work with corpora, use complex search expressions, perform part-of-speech annotation and dependency parsing, and much more – all from within R.
- The `ggplot2` package offers an incredibly powerful framework for data visualisation. Not convinced? Check out the ggplot2 gallery.

- The CRAN repository features **more than 20,000 packages** that can be installed to expand the functionality of R almost indefinitely. Should none of them meet your needs, R gives you the tools to **write and share your own functions and packages** with ease.

Installing R

The first step involves downloading the R programming language itself. The link will take you to the homepage of the Comprehensive R Archive Network (CRAN), where you can download the binary distribution. Choose the one that corresponds to your operating system (Windows, macOS or Linux).

> **i** Installation instructions for Windows users
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> Click "Download R for Windows" → Select "base" → Click on "Download R-4.4.1 for Windows" (or whichever version is currently displayed).
> Open the setup file you've just downloaded and simply follow the on-screen instructions. It's fine to go with the default options.
> Video tutorial on YouTube

> **i** Installation instructions for macOS users
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> Click "Download R for macOS" → Select the latest release for your Mac. Note that there are separate versions for Macs with Apple silicon (M1, M2, etc.) and for older Macs with Intel processors.
> Open the downloaded .pkg file and follow the instructions in the installation window.
> Video tutorial on YouTube

Installing RStudio

You can now download and install RStudio. RStudio is a so-called 'integrated development environment' (IDE), which provides us with a variety of helpful tools for writing and editing code comfortably. If R were a musical instrument, RStudio would be the recording studio, so to speak.